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Method and apparatus for bacterial analysis

Abstract

An apparatus (10) comprising: a light source (12); to cast light toward a substrate (20) defining a bacteria binding volume to create an evanescent field (22), the bacteria binding volume being within the evanescent field; a detector (32, 34) arranged to receive light from the bacteria binding volume and output data (36, 37); and a processor (38) arranged to determine vibration of bacteria (26) with the bacteria binding volume in three-dimensions from the data.

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Background/Summary

(1) This application is a U.S. national phase application under 35 U.S.C. § 371 of International Application No. PCT/GB2018/052154, filed Jul. 30, 2018, which claims priority to U.K. Application No. 1712279.7 filed Jul. 31, 2017, the complete disclosure of which is incorporated

herein by reference.

BACKGROUND OF THE INVENTION

(2) Recently, the rise in antimicrobial resistance has become problematic for the treatment of bacterial infections. There is therefore a need for antimicrobial resistance to be detected more rapidly and at lower cost.

SUMMARY OF THE INVENTION

(3) A first aspect of the invention provides an apparatus comprising: a light source arranged to cast light toward a substrate defining a bacteria binding volume to create an evanescent field and the bacteria binding volume is within the evanescent field; a detector arranged to receive light from the bacteria binding volume and output data; and a processor arranged to determine a level of vibration of bacteria within the bacteria binding volume in three-dimensions from the data.

(4) With such an arrangement, there is provided a simpler apparatus for detecting antimicrobial resistance.

(5) The light from the bacteria binding volume can be any light which contains information regarding the level of vibration of the bacteria. This can be light which has been scattered by bacteria due to their movement in the evanescent field and/or can be light which has been totally internally reflected and so has the profile of the incident light except that the light lacks the photons which are in the scattered light.

(6) The light source may be a laser light source. With such an arrangement, the evanescent field can be created more reliably.

(7) The apparatus may further comprise a lens arranged to direct the light from the light source towards the substrate. With such an arrangement, the evanescent field can be produced by light arriving at an angle closer to the critical angle.

(8) The lens may be a high numerical aperture lens, having a numerical aperture of 1.4 or more, such as 1.41. With such an embodiment, the evanescent field can be produced with a sufficiently large size.

(9) The lens may be arranged to receive light from the bacteria binding volume. With such an arrangement, the light from the bacteria binding volume can be directed toward the detector without the use of a greater number of lenses.

(10) The detector may be a quadrant photodiode detector. With such an arrangement, the movement of the bacteria in four different sectors can be determined in order that the vibration of the bacteria in the binding volume can be measured more precisely.

(11) The determined vibration may be of a single bacterium. With such an arrangement, no culturing of bacteria is required so that the overall process of determining antimicrobial resistance can be made more rapidly.

(12) The apparatus may further comprise a coverslip arranged to support the bacteria binding volume. With such an arrangement, there is provided a transparent support for the bacteria binding volume so that the evanescent field may be created without light from the light source travelling through the bacteria binding volume.

(13) The light source and lens may be arranged to cause the light to undergo total internal reflection on the surface of the coverslip. With such an arrangement, the evanescent field can be correctly positioned to contain bacteria bound by the bacteria binding volume.

(14) The evanescent field may be created on the surface of the coverslip. With such an arrangement, the evanescent field is correctly positioned to be affected by vibrations of bacteria bound within the bacteria binding volume.

(15) The evanescent field may have a decay length of no more than two hundred nanometres. With such an arrangement, the evanescent field will not extend far past the bacteria, in order that the scattering of light can be caused by the bacteria only.

(16) The first detector may be arranged to receive light scattered by the bacteria in the binding volume. With such an arrangement, the bacteria can be analysed with less processing.

- (17) The apparatus may further comprise a second detector arranged to receive light from the light source having undergone total internal reflection, which is not scattered by the bacteria. With such an arrangement, the second detector can provide a check in order to determine whether the data received from the first detector is correct and may be used to measure the fluctuations of the bacteria.
- (18) The apparatus may further comprise a second light source, the second light source being arranged to direct light through the coverslip, the light from the second light source undergoing multiple internal reflections within the coverslip. With such an arrangement, the vibrations of multiple bacteria can be detected in order to give a determination of antimicrobial resistance with the apparatus in use for a shorter time.
- (19) The coverslip may be formed of a dielectric material having a refractive index greater than 1.6. With such an arrangement, the total internal reflections can be reliably produced.
- (20) The apparatus may further comprise a second light source and an optical waveguide arranged to support the bacteria binding volume, the second light source arranged to direct light through the optical waveguide. With such an arrangement, there is provided an alternative arrangement without use of a coverslip.
- (21) The optical waveguide may be tapered at a point within the evanescent field. With such an arrangement, the apparatus can be more sensitive to the vibrations of the bacteria within the evanescent field, having a greater signal-to-noise ratio.
- (22) The apparatus may further comprise a CMOS or CCD imaging device arranged to observe the bacteria. With such an arrangement, a manual measurement of the level of vibration can be made in order to check an automated result.
- (23) The vibrations within each bacterium may be recorded and stored as video footage. The footage may have a high resolution such that an observer can visually analyse the vibrations. With such an arrangement, the video can be analysed to give a measurement of bacterial activity alternatively or in addition to the measurement obtained from the position sensitive detector.
- (24) A further aspect of the invention provides a method comprising: arranging one or more bacteria on a surface of a coverslip; directing light to be totally internally reflected on the surface of the coverslip and thereby creating an evanescent field on the surface of the coverslip, the evanescent field containing the bacteria; detecting light from the evanescent field; and determining a level of vibration of the bacteria from the detected light.
- (25) The detected light can be any light which contains information regarding the level of vibration of the bacteria. This can either be light which has been scattered by bacteria due to their movement in the evanescent field or can be light which has been totally internally reflected and so has the profile of the incident light except that the light lacks the photons which are in the scattered light.
- (26) With such an arrangement, there is provided a more simple method for determining antimicrobial resistance. The determining may comprise determining movement of the bacteria in three dimensions. With such an arrangement, a more accurate picture of the movement of the bacteria can be obtained.
- (27) The determining may comprise determining the movement of individual bacteria. With such an arrangement, there is no requirement for culturing and so the overall time required for the determination of antimicrobial resistance can be reduced.
- (28) The arranging may comprise absorbing the bacteria to the coverslip. In such an arrangement, the bacteria can be properly constrained to remain within the evanescent field.
- (29) The method may further comprise determining the life status of the bacteria from the vibrations. With such an arrangement, a more complete determination regarding antimicrobial resistance can be made.
- (30) The light may be laser light. With such an arrangement, the evanescent field can be reliably produced.
- (31) The light may be directed toward the coverslip by a lens and the lens is also arranged to

receive light from the coverslip. With such an arrangement, the light can be properly directed on the coverslip and directed toward a detector without the use of two lenses.

(32) The detector may be a quadrant photodiode detector. With such an arrangement, the movement of the bacteria between four different quadrants can be detected in order to give a complete picture of the vibration of the bacteria.

(33) The evanescent field may have a decay length of no more than two hundred nanometres. With such an arrangement, the evanescent field is appropriately sized such that light is not scattered by elements other than bacteria.

(34) The detected light may be light which has been scattered by the bacteria in the bacteria binding volume. With such an arrangement, the bacteria can be analysed with less processing of the data.

(35) The method may further comprise detecting light from the light source, which has undergone total internal reflection and is not scattered by the bacteria, with a second detector. With such an arrangement, there is provided a system for checking the data produced by the first detector in order to provide a more reliable result.

(36) The method may further comprise directing light from a second light source through the coverslip, the light from the second light source undergoing multiple total internal reflections within the coverslip. With such an arrangement, scatterings from multiple bacteria can be detected in order to give a determination of antimicrobial resistance with less time spent using the apparatus.

(37) The method may further comprise directing light from a second light source through an optical waveguide, the light from the light source undergoing multiple total internal reflections within the optical waveguide. With such an arrangement, there is provided an alternative method which does not require the use of a coverslip.

(38) The optical waveguide may be tapered at a point within the evanescent field. With such an embodiment, the apparatus can be more sensitive to the vibration of bacteria in order to give a higher sensitivity for determining antimicrobial resistance.

(39) The determining may be carried out by a processor. With such an arrangement, the determination can be carried out more quickly.

(40) The method may further comprise adding an antimicrobial to the coverslip. With such an arrangement, a determination regarding antimicrobial susceptibility may be made.

(41) The arranging of bacteria on the coverslip may further comprise contacting a portion of the bacteria with an antimicrobial and having a portion of the bacteria uncontacted by the antimicrobial. With such an arrangement, a determination regarding antimicrobial resistance may be made more quickly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention will now be described with reference to the accompanying drawings, in which:

(2) FIG. 1 shows an overall arrangement in accordance with an embodiment of the invention.

(3) FIG. 2 shows a detailed drawing of a coverslip or waveguide arrangement in accordance with an alternative embodiment of the invention.

(4) FIG. 3 shows a detailed drawing of a tapered waveguide arrangement in an alternative embodiment of the invention.

DETAILED DESCRIPTION OF EMBODIMENT(S)

(5) FIG. 1 shows an overall system for observing vibrations of bacteria **26** in order to determine antimicrobial resistance.

(6) In an antimicrobial detection system **10**, there is a light source **12**, which is preferably a laser

light source, for example having a wavelength of 561 nanometres. The light source **12** produces a light beam **14** which is then refracted by a lens **16**. The lens **16** may be a high numerical aperture lens, combined with immersion oil **40** which refracts the light beam **14** such that the light hits the top surface of a coverslip **20** at an angle greater than the critical angle and is totally internally reflected. It will be understood by a person skilled in the art that the critical angle is a function of the refractive indices of the coverslip **20** and the medium above the coverslip **20**.

(7) While the lens **16** having a numerical aperture equal to that of the medium above the lens **16** (approximately 1.33 if the medium is water) would result in total internal reflection and the creation of an evanescent field, the lens **16** having a numerical aperture of 1.4 or greater can produce an evanescent field of appropriate size for the analysis of bacteria.

(8) The coverslip **20** can be a commercially available thin glass coverslip. The coverslip **20** may also be provided separately from the rest of the system **10** and exchangeable for different coverslips having different bacteria and different thicknesses as required by each user. While the coverslip **20** might not be part of the system **10**, the system **10** includes a coverslip support **18**. The system **10** can therefore be arranged such that the system **10** will function correctly when the coverslip **20** is positioned on the coverslip support **18**, without the coverslip **20** being present at the time of the arranging.

(9) The light **14** incident on the top surface of the coverslip **20** forms an evanescent field **22** which forms a circle on the coverslip surface with a Gaussian intensity distribution and a maximum decay length of approximately 200 nanometres. Other configurations are possible for the evanescent field **22**; for example the field may be elliptical and have a non- Gaussian intensity distribution. The evanescent field **22** interacts with objects on the surface of the coverslip such as bacteria **26**.

(10) Bacteria **26** are elastically bound to the coverslip **20** using antibodies **24** such that the bacteria remain roughly in the centre of the evanescent field **22**. The coverslip **20** can be prepared by first being cleaned using ethanol then distilled water and dried in a stream of nitrogen. The surface is then amino-functionalised using ethanolamine hydrochloride, which results in a covering of amino groups covalently bound to the glass to which antibodies **24** can bind. The antibodies **24** used should be those specifically targeted to bind the subject bacteria **26**.

(11) The coverslip **20** may form a substrate and the antibodies **24** on the coverslip **20** may form a bacteria binding volume for binding the bacteria **26** in place.

(12) In use, light which has been scattered by the bacteria is then transmitted from the coverslip as scattered light **28**. The light passes through the lens **16** and into a first detector **32**, which may be a quadrant photodiode detector, which then produces data **36** to provide to a computer system **38**.

(13) A quadrant photodetector can monitor both light intensity and the position of peak intensity. Therefore, a quadrant photodetector can be used to determine the distance from the coverslip **22** to the bacteria **26** by the magnitude of the peak intensity of the light detected. The position of the bacteria **26** within the plane of the coverslip **22** can then be determined by the position of the peak intensity detected. Alternatively, an array of photodetectors such as CMOS or CCD may be used in order to determine the position of the bacteria.

(14) Light which has been reflected by the top surface of the coverslip **20** is transmitted from the coverslip as un-scattered light beam **30**, which, optionally or alternatively, also passes through the lens **16** and to a second detector **34**. The second detector **34** produces second data **37** which are transferred to a computer system such as the computer system **38**. As will be understood to the skilled person, either the first detector **32** or the second detector **34** can be omitted from the system or both may be used as required. The second detector **34** can be a quadrant photodetector identical to that of the first detector **32**, or can be an array of photodetectors.

(15) In order to direct the light from the light source **12** to the coverslip **20** and from the coverslip **20** to the photodetectors **32**, **34**, the system can comprise mirrors **39**.

(16) The computer system **38** can contain a non-transient computer readable medium and a processor, the non-transient computer readable medium containing instructions for the analysis of

the data **36, 37** by the processor in order to provide results to a technician for analysis.

(17) When using the system **10**, the bacteria **26** can be arranged on the coverslip **20** using the antibodies **24** and the vibrations of the bacteria **26** can be recorded and stored by the computer system **38**. An antimicrobial can then be added to the surface of the coverslip **20** and the subsequent vibration of the bacteria **26** on the coverslip **20** can be recorded by the computer system **38**.

(18) If the bacteria **26** are arranged in a liquid environment then the antimicrobial can be added to the liquid environment, either partially or totally. If the antimicrobial is added only to a part of the environment, then the bacteria **26** can be separated spatially on the coverslip **20** so that a group of bacteria **26** uncontacted by the antimicrobial and a group of bacteria **26** contacted by the antimicrobial can be analysed at the same time. Alternatively, a group of bacteria **26** uncontacted by the antimicrobial can be present before the addition of the antimicrobial and the same bacteria **26** can be observed after the addition of the antimicrobial.

(19) If the two groups of bacteria **26** are separated in space (analysed at the same time), then a time saving can be made, whereas if the two groups are separated in time then a greater sample size may be used.

(20) Once the data **36, 37** has been gathered by the computer system **38** it can be processed. In the processing, the level of activity of the bacteria **26** after the addition of the antimicrobial can be compared to the level of activity of the bacteria **26** before the addition of the antimicrobial.

(21) If the level of activity of a bacterium has increased significantly, then it can be assumed that the bacterium has become unbound from the antibody **24** and the result can be discarded. If the level of activity has remained approximately constant, then the bacterium can be considered not to have been affected by the antimicrobial, and if the bacterium has significantly reduced in activity then the bacterium can be deemed to have been affected by the antimicrobial. Antimicrobials can be bactericides or bacteriostats.

(22) The determination of whether a change in a level of activity is significant may be determined by standard statistical testing, such as requiring the change to be greater than 5% to be significant and otherwise determining that the level is approximately unchanged.

(23) The above-analysis can be repeated until a determination can be reached as to whether the bacteria exhibit antimicrobial resistance or not.

(24) FIG. 2 shows a zoomed in image of a coverslip or waveguide arrangement **100** including a coverslip **116**, which is functioning as an elongated waveguide. In order to function as a waveguide, the coverslip **116** can have a refractive index greater than 1.6, and optionally be a sapphire coverslip. The coverslip will then function as a waveguide in the apparatus **10** of FIG. 1, in particular in the presence of immersion oil **40**.

(25) A second light source **112** provides light **114** which enters coverslip **116** and undergoes multiple total internal reflections. When undergoing the multiple total internal reflections, light **114** creates evanescent field **118**, which contains bacteria **122** bound to the coverslip by antibodies **120**. As above, the activity of the bacteria **122** within the evanescent field **118** results in scattering of light **114**. Light leaves the coverslip **116** as a light beam **115**, which is affected by the bacteria **122**. A light detector **124** receives the light beam **115**.

(26) The second light source **112** can be a laser light source the same as the first light source **12**, and the light detector **124** can be a quadrant photodetector or one or more of CMOS OR CCD detectors, which may have the same properties as the first and second detectors **32, 34**.

Alternatively, a more simple detector measuring only light intensity but not position can be used.

(27) The coverslip arrangement **100** shown in FIG. 2 can be combined with the system **10** of FIG. 1 or may be used on its own, as required. If the coverslip arrangement **100** of FIG. 2 is combined with the system **10** of FIG. 1, then light scattered by the bacteria **122** can be detected by the light detectors of FIG. 1.

(28) In use, the evanescent field produced by the light beam **114** in the coverslip **116** of the coverslip arrangement **100** can encompass multiple bacteria such that light is scattered by more

than a single bacterium. Hence, the light beam **115** which exits the coverslip **116** and is detected by light detector **124** contains data produced by interactions related to multiple bacteria **122**.

(29) A tapered optical waveguide **216** is shown in FIG. **3**. The system **200** of FIG. **3** is similar to the system **100** of FIG. **2**. In this system, a light source **212** provides a light beam **214** which undergoes multiple total internal reflections within the optical waveguide **216** and creates an evanescent field **218** containing bacteria **222** bound to the optical fibre **216** by antibodies **220**. The vibrations of the bacteria **222** results in changes to the light beam **214**, which results in a different light beam **215** exiting the optical fibre **216**, which is detected by detector **224**.

(30) The tapered waveguide **216** differs from the coverslip **116** due to the tapered sides of the tapered waveguide **216**. The tapering of the tapered waveguide **216** results in the waveguide being thinner in the middle, such that the overall system can be more sensitive to vibrations within the evanescent field **218**. Although the tapering is shown on both sides of the tapered waveguide **216**, the tapering may take place on only one side or on both sides. If the tapering takes place on only one side, then this side may be either the top side or the bottom side.

(31) The waveguide **216** should not have any cladding on the outside, as this enables the evanescent field **218** to be exposed such that the bacteria **222** can be arranged within the evanescent field.

(32) The processing for the coverslip arrangements **100** and **200** may be substantially similar and different from the processing for the system **10**. In the processing of the data produced from the coverslip arrangements **100** and **200**, each dataset contains details of multiple bacteria and aggregated vibrations. The analysis therefore takes a broader overview, with activity calculated for sets of bacteria rather than for individual bacteria. There is therefore less need for repeat readings with the data produced by the coverslip arrangements **100** and **200**.

(33) Where the word ‘or’ appears this is to be construed to mean ‘and/or’ such that items referred to are not necessarily mutually exclusive and may be used in any appropriate combination.

(34) Although the invention has been described above with reference to one or more preferred embodiments, it will be appreciated that various changes or modifications may be made without departing from the scope of the invention as defined in the appended claims.

Claims

1. An apparatus comprising: a light source arranged to cast light toward a transparent coverslip supporting a bacteria binding volume to create an evanescent field, the bacteria binding volume being within the evanescent field; a detector arranged to receive light from the bacteria binding volume and output data; a processor programmed to determine a level of vibration of bacteria within the bacteria binding volume in three-dimensions from the output data; the transparent coverslip arranged to support the bacteria binding volume; the bacteria binding volume; and a coverslip support arranged to support the coverslip, the coverslip support being arranged between the light source and the coverslip and having a hole to allow the light cast from the light source to pass to the transparent coverslip; wherein the evanescent field is created on a surface of the transparent coverslip, wherein the bacteria binding volume is supported directly on the transparent coverslip; and wherein the transparent coverslip has no cladding.
2. The apparatus of claim 1, further comprising a lens arranged to direct the light from the light source toward the coverslip.
3. The apparatus of claim 2, wherein the lens has a numerical aperture of at least 1.4.
4. The apparatus of claim 2, wherein the lens is arranged to receive light from the bacteria binding volume.
5. The apparatus of claim 1, wherein the determined vibration is of a single bacterium.
6. The apparatus of claim 1, wherein the detector is arranged to receive light scattered from the bacteria within the bacteria binding volume.

7. The apparatus of claim 6, wherein the apparatus further comprises a second detector, arranged to receive light from the light source having undergone total internal reflection, which is not scattered by the bacteria.
 8. The apparatus of claim 1, further comprising a second light source, the second light source being arranged to direct light through the coverslip, the light from the second light source undergoing multiple total internal reflections within the coverslip.
 9. The apparatus of claim 8, wherein the coverslip is formed from a dielectric material having a refractive index greater than 1.6.
 10. The apparatus of claim 1, further comprising a second light source and wherein the transparent coverslip is arranged to act as an optical waveguide arranged to support the bacteria binding volume, the second light source being arranged to direct light through the optical waveguide.
 11. The apparatus of claim 10, wherein the optical waveguide is tapered at a point within the evanescent field.
 12. The apparatus of claim 1, wherein the coverslip is separable from the coverslip support.
 13. The apparatus of claim 1, wherein the surface of the coverslip where the evanescent field is created contacts the bacteria binding volume.
 14. The apparatus of claim 1, wherein the evanescent field is produced by total internal reflection at, at least a surface of the coverslip which contacts the bacteria binding volume.
 15. The apparatus of claim 1, wherein the coverslip support directly contacts the coverslip.
 16. The apparatus of claim 1, wherein the coverslip is a stand-alone, solid material.
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